

Gaussian Processes Part 2

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Lecture 5



① Introduction

② GP approach

- Introduction
- Non-periodic part
- Periodic part
- The Model
- Fourier Features

③ Example

- Stochastic Volatility



Time Series, Classical Approach

If data has seasonality, you usually use **STL** decomposition. However,

- Parameters are not interpretable, only decomposition is available
- No uncertainty estimates
- Quite strict on input values
- Significantly less flexible in modelling

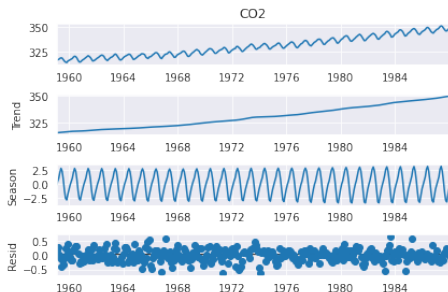


Figure: STL decomposition for CO2 data, Statsmodels



GP decomposition

A Gaussian process can handle a complicated set of assumptions in addition to what STL provides

- Granular seasonality (year + quarter + month + week)
- Changepoint models
- Flexible likelihood Function
- Panel regression models
- Missing values



Typical Model

Typical model is additive

$$x_t \sim \underbrace{g(t)}_{\text{non-periodic}} + \underbrace{s(t)}_{\text{periodic}} + \underbrace{h(t)}_{\text{holidays}}$$

Reference

See more in **Prophet** preprint [2]. Every time series model is unique



Reminder

$$x \in \mathbb{R}^n, y \in \mathbb{R}$$

$$Y \sim \mathcal{GP}(m(x), k(x, x'))$$



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- ① $\mathcal{GP}$ Gaussian Process - simply, a normal distribution with special mean $m(x)$ and covariance $k(x, x')$



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 - Linear regression $m(x) = x^\top \beta$
 - Simply Constant or Zero $m(x) = c$
 - Other custom functions $m(x) = \sin(x)$



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 - Simply Constant or Zero $m(x) = c$
 - Other custom functions $m(x) = \sin(x)$
- ③ $k(x, x')$ - kernel function, simply - measure of similarity for x and x'
 - $[K]_{ij} = k(x_i, x_j)$ is an SPD matrix



Non-periodic Part (mean function)

- Growth models
- Linear trend models
- Changepoint models



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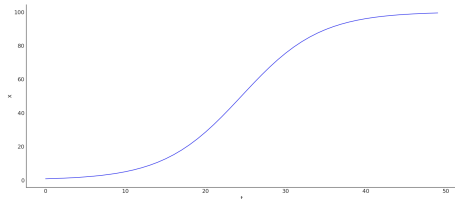


Figure: Growth Model

$$x = \frac{c}{1 + \exp(-k(t - m))}$$



Non-periodic Part (mean function)

- Growth models
- **Linear trend models**
- Changepoint models

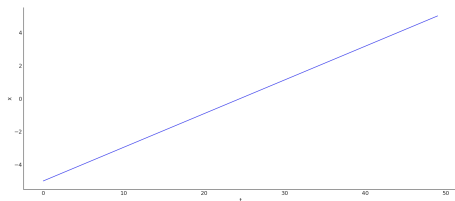


Figure: Linear Trend Model

$$x = \frac{c}{1 + \exp(-k(t - m))}$$



Non-periodic Part (mean function)

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- Linear trend models
- **Changepoint models**

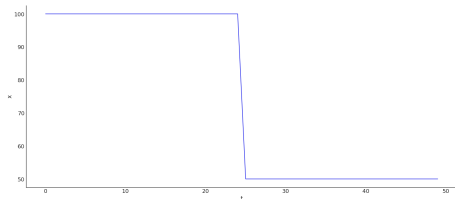


Figure: Changepoint Model

$$x = \begin{cases} c_1, & t < m \\ c_2, & t \geq m \end{cases}$$



Non-periodic Part (mean function)

- Growth models
- Linear trend models
- Changepoint models

Extentions

Extensions are possible, e.g. time dependent saturation in the growth model. See in [2]

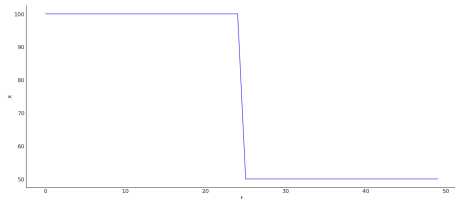


Figure: Changepoint Model

$$x = \begin{cases} c_1, & t < m \\ c_2, & t \geq m \end{cases}$$



Holidays

$$h(t) = \text{is-holiday}(t)$$

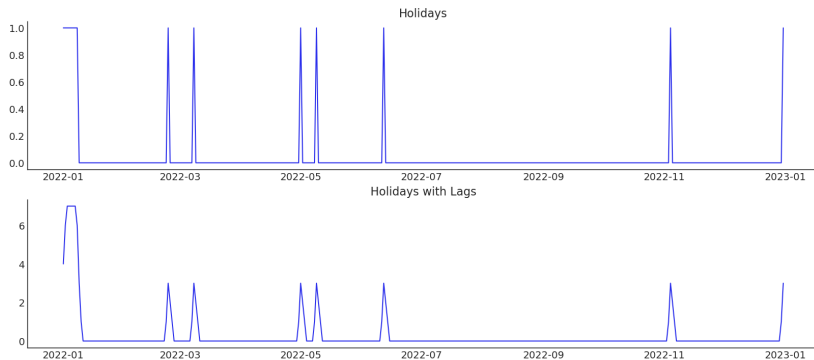


Figure: Holiday features



Periodic part (cov function)

Granularities are important here.
Multiple Periodic kernels can be used.

- Yearly
- Quarterly
- Monthly
- Weekly

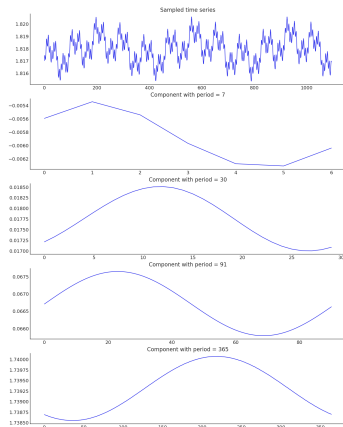


Figure: Seasonal Decomposition



Lengthscales for Periodic Part

Hyperparameters

Common sense driven lengthscale choice

- Week - couple of days make a change ($l_s \approx 3$)
- Month - week makes sense for a dramatic change ($l_s \approx 7$)
- Quarter - month makes sense for a dramatic change ($l_s \approx 30$)
- Year - quarter makes sense for a dramatic change ($l_s \approx 90$)



Lengthscales for Periodic Part

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Common sense driven lenthscale choice

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- Month - week makes sense for a dramatic change ($l_s \approx 7$)
- Quarter - month makes sense for a dramatic change ($l_s \approx 30$)
- Year - quarter makes sense for a dramatic change ($l_s \approx 90$)

In practice

Everybody is using Fourier features as a replacement for Periodic Kernel



Putting All Together

$$m(t) = \underbrace{g(t)}_{\text{non-periodic}} + \underbrace{h(t)}_{\text{holidays}}$$

$$\begin{aligned} k(*, *) = & \text{Periodic}(p = 365, l = 90) \\ & + \text{Periodic}(p = 90, l = 30) \\ & + \text{Periodic}(p = 30, l = 7) \\ & + \text{Periodic}(p = 7, l = 3) \end{aligned}$$



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Missing Parts

- 1 Weights for periodic components



Putting All Together

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Missing Parts

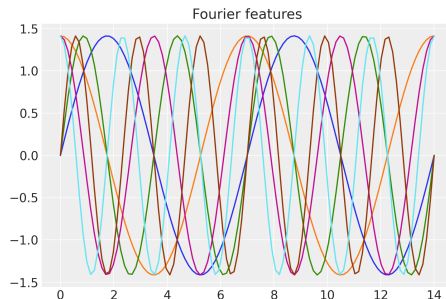
- 1 Weights for periodic components
- 2 Trend violations from $g(t)$



More Efficient Periodic

Having at least one Periodic kernel in a large time series prevents from optimizations.

- Fourier features can be added as regressors
- This allows reasonable periodicity to be present in the model





Choosing Features

Every regular period has optimal number of fourier components (aka order)

- Weekly: 3
- Monthly: 10
- Yearly: 5

```
from collections import namedtuple
from enum import Enum

Season = namedtuple("Season", "period,order")

class Daily(Season, Enum):
    Week = 7.0, 3
    Year = 365.25, 5
    Month = 365.25 / 12, 10

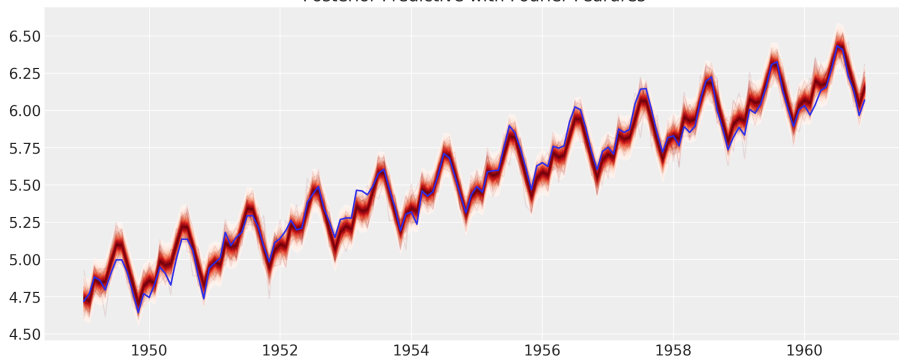
class Monthly(Season, Enum):
    Year = 12, 5
    Month = 1, 10
```



Applications of Fourier Features

- E.g. Prophet uses Fourier Features (FF)
- It is VERY Fast, drop-in replacement for Periodic GP
- Proven to be useful for Bayesian models

Posterior Predictive with Fourier Features





Integrating into the model

```
alpha = pm.Normal("alpha", 0, sigma, shape=Monthly.Year.order * 2)
features = fourier_series(months, season=Monthly.Year)
seasonality = features @ alpha

expectation = bias + trend + seasonality
```



Stochastic Volatility Model

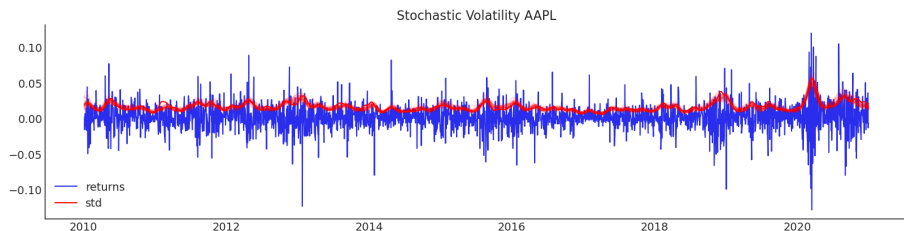


Figure: Stochastic Volatility Estimation for AAPL

Read More

<https://github.com/quantopian/bayesalpha>



Key Ideas

- Returns are constant
- Volatility is not constant
- Model volatility as a Gaussian process
- Use approximations to speed up the model
- Use GPU to do fast inference



Priors

- Returns
- Volatility
- Time Component



Priors

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 - Expect year return at orders $\pm 100\%$
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 - $\log \text{std} = \log(2^{\frac{1}{250}} - 1)$
- Time Component
 - After the model is framed



The Model

I did the modelling choice to make it as simple as possible

$$\text{return} \sim \text{Normal}(0, 2^{\frac{1}{250}} - 1)$$

$$\text{log std} \sim \text{Normal}(\log(2^{\frac{1}{250}} - 1), 0.05)$$

$$\text{ls} \sim \text{Gamma}(30, 5)$$

$$\alpha_{\text{vol}} \sim \text{Exponential}(100)$$

$$K(*, *) = \alpha_{\text{vol}} \text{Matern32}(\text{ls})$$

$$\Delta_t^{\text{log std}} \sim \mathcal{GP}(0, K)$$

$$\text{obs}_t \sim \text{Normal}(\text{return}, \exp(\text{log std} + \Delta_t^{\text{log std}}))$$



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Make the model faster!

HSGP



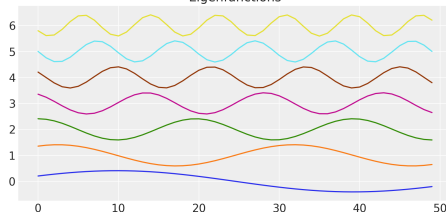
What is HSGP?

HSGP [1] is a short of **H**ilbert **S**pace **G**aussian **P**rocess.

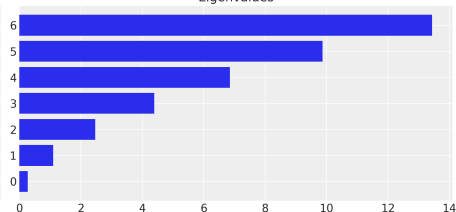
Gentlemen FAQ:

- Approximates in function space
- Works for Stationary kernels
- Periodic Kernels are yet tricky
- Implemented in PyMC

Eigenfunctions



Eigenvalues





Why HSGP?

- ① Time series models have a lot of time-points
- ② Algorithm complexity grows very fast ($\mathcal{O}(n^3)$)
- ③ HSGP [1] reduces the complexity to just $\mathcal{O}(mn + m)$
 - m - number of basis functions)

Gaussian process are now fast!

From $\mathcal{O}(n^3)$ to $\mathcal{O}(mn + m)$ makes so much sense!



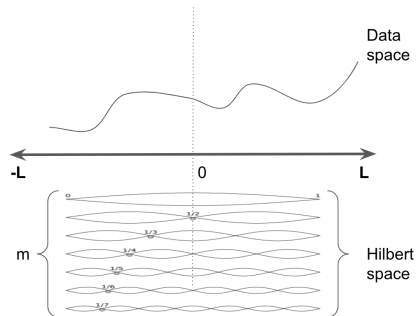
How to HSGP

There are two main parameters:

- L - extrapolation boundary
- m - number of basis functions

You can use it

No need to be a mathematician to set L or m , there are good defaults in documentation.





HSGP for Stochastic volatility

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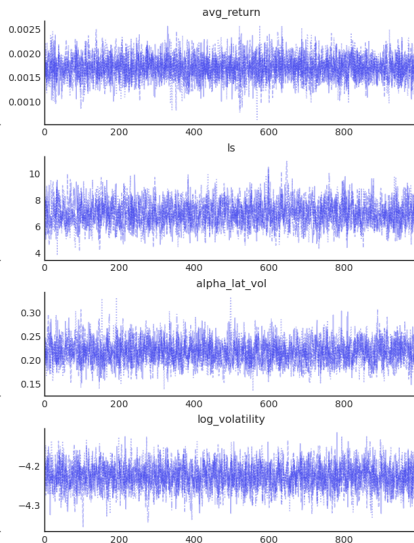
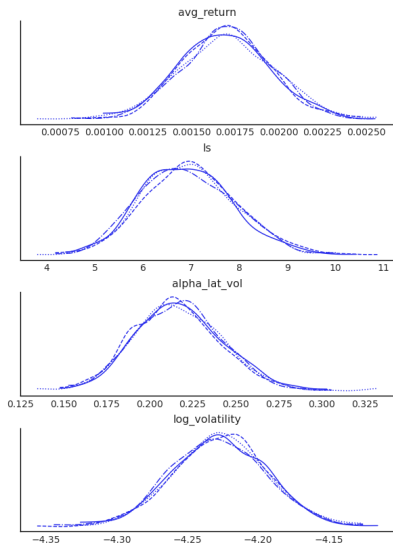
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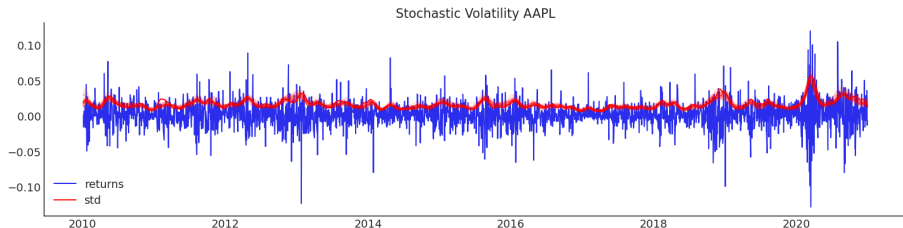


Results





Takeaways



- Interpretable parameters
- Splines make model much faster
- Uncertainties for
 - average volatility
 - average returns
 - stochastic variation (α_{vol})



References I



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